SLR-GB-84

Seat No.	

Set P

B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination: October/November – 2025
Data Structures using C++ (G06-DSC1-0301)

Day & Date: Thursday, 30-10-2025 Max. Marks: 30
Time: 12:00 PM To 01:30 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternatives. **06**

1) The operation which is processing each element in a list is known as _____.
a) Sorting b) Merging
c) Inserting d) Traversing

2) A data structure in which elements can be inserted or deleted at/from both ends but not in the middle is _____.
a) _Queue b) Circular queue
c) Dequeue d) Priority queue

3) The type of expression in which operator succeeds its operands is _____.
a) Infix Expression b) Prefix Expression
c) Postfix Expression d) Both Prefix and Postfix Expressions

4) Which of these is not an application of a linked list?
a) To implement file systems
b) For separate chaining in hash-tables
c) To implement non-binary trees
d) Random Access of elements

5) Which of the following traversal strategy not used in binary tree?
a) Preorder traversal b) Inorder traversal
c) Postorder traversal d) Priority traversal

6) Which of the following concepts make extensive use of arrays?
a) Binary trees b) Scheduling of processes
c) Caching d) Expression validity

Q.2 Answer the following. (Any Three) **06**

a) State the applications of stack.
b) What is Array? Write syntax to initialize one dimensional array.
c) List out different applications of queue.
d) What is searching? How binary search is better that linear search?
e) Define the term: Complete Binary tree.

Q.3 Answer the following. (Any Two) **06**

a) Explain ADT of array.
b) Differentiate between stack and queue.
c) Explain reverse( ) operations of Circular Doubly Linked List with diagram.

Q.4 Answer the following (Any Two) **06**

a) What do you mean by Heap tree? Explain types of Heap tree.
b) Write a menu driven program to implement singly linear linked list with following operations -
1) Insert node at end.
2) Search node.
3) Count nodes.

c) Write an algorithm for binary search method.

Q.5 Answer the following. (Any One) **06**

a) Write a C++ program to implement Queue using arrays.
b) Write an algorithm of merge sort. And apply Merge Sort to sort given Series in descending order: 4, 9, 105, 74, 10, 85, 71, 101, 876, 143,39, 571,24.